

Governed Execution Framework (GEF)

Deterministic Supervision

and

Execution Governance in Probabilistic AI Systems

Executive Summary: Highlighting the shift from additive prompt-wrapper guardrails to structural, deterministic supervision.

The Ambient Authority Problem: Explaining why probabilistic LLMs cannot hold execution keys or ambient capabilities.

Architectural Decoupling: Defining the strict boundary between reasoning proposals and execution authorization.

Deterministic Supervision Boundaries: Explaining the supervisor gate, hard operational limits, and zero ambient authority.

Auditable Traceability: Explaining forensic logging and legal defensibility through immutable execution artifacts.